Cairo University – Faculty of Engineering

Credit Hours System – Spring 2025

SBES161 – Medical Monitors and Life Support Equipment

Task 3

Endoscopic Image Enhancement System

Prepared By:

Farah Ahmed Kamal 4230162

Salma Mohamed Saad 4230195

Menna Hesham Allam 1220321

Maryam Mohamed Ali 1220315

Presented To:

Eng. Abdulrehman Suliman

Objective:

The goal of this task was to design and implement a basic endoscopic system prototype that includes mechanical articulation, illumination control, and real-time image capture and transmission, mimicking functionalities of a medical endoscope.

System Components and Implementation

1. Mechanical Control (Articulation System)

To enable directional control of the endoscope, we used two servo motors to simulate X- and Y-axis movement.

These were controlled by an Arduino Uno and navigated using a joystick module, allowing intuitive manual positioning of the endoscopic shaft.

The servos were attached to pins 3 and 5.

- **Joystick Input:**
  - X-axis: A2
  - Y-axis: A3
- **Servo Angles:** Mapped from 0–1023 (analog input) to 0–180° for realistic articulation.

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2. Illumination System

The lighting module included both white and RGB LEDs:

- **White LED Brightness Control:** A potentiometer connected to pin A1 adjusted the brightness of 4 white LEDs connected to pin 6.
  This simulated variable illumination intensity during endoscopic procedures.

- **Color Temperature Simulation (RGB Control):** Another potentiometer (A0) was used to smoothly cycle through different color temperatures by adjusting the RGB LED values (connected to pins 9, 10, and 11).
  A custom mapping was implemented to transition across Red → Green → Blue, simulating changing lighting environments.

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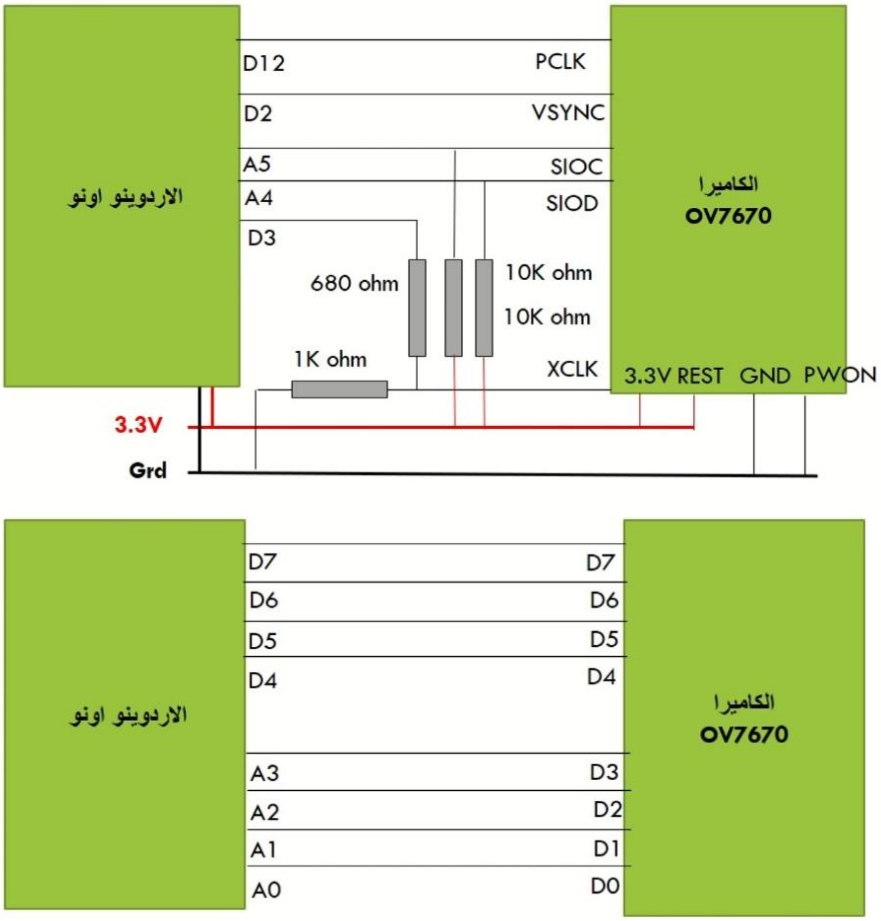
3. Image Capture System

An OV7670 camera module was used for real-time image acquisition. The system was designed to capture different frames, the camera interfacing was integrated as a separate Arduino due to its specific memory and timing requirements.

Wiring Overview

1. Camera OV7670



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2. Servo Motors (X-axis and Y-axis Control)

- **Servo 1 (X-axis)**: Signal wire connected to **Digital Pin 3**
- **Servo 2 (Y-axis)**: Signal wire connected to **Digital Pin 5**
- Both servos are powered through **5V** and **GND** pins on the Arduino

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3. *Joystick Module*

- **X-axis output (VRx)** connected to **Analog Pin A2**
- **Y-axis output (VRy)** connected to **Analog Pin A3**
- The joystick's **VCC** is connected to **5V**, and **GND** to **GND**

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4. *White LED (Brightness Controlled)*

- The **positive leg (anode)** of the white LED is connected directly to **Digital Pin 6**
- The **negative leg (cathode)** is connected to **GND**
- LED brightness is controlled via a **potentiometer** connected as follows:
  - **Middle pin** to **Analog Pin A1**
  - **Side pins** to **5V** and **GND**

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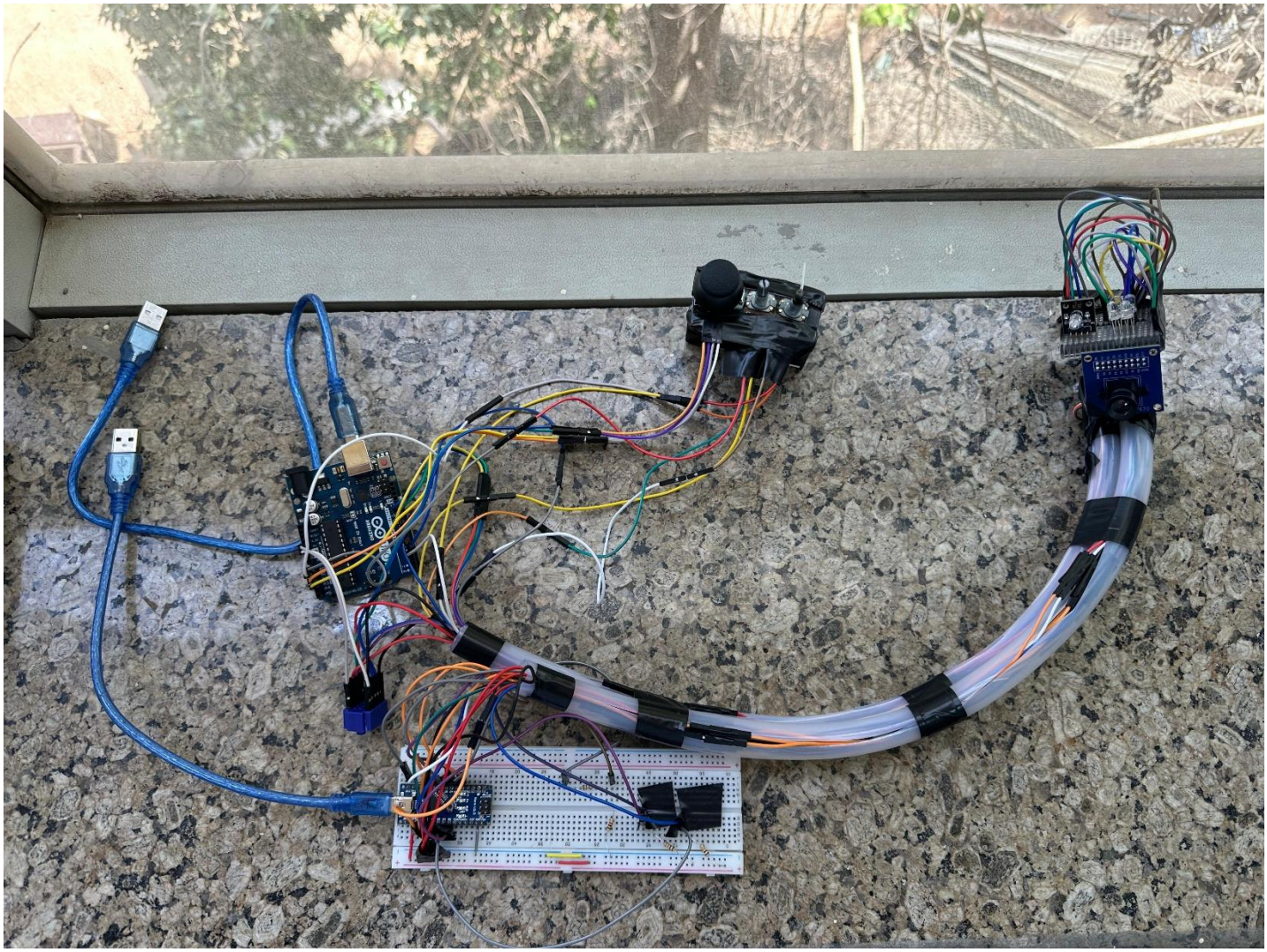
5. *RGB LED (Color Controlled)*

- **Red pin** connected to **Digital Pin 9**
- **Green pin** connected to **Digital Pin 10**
- **Blue pin** connected to **Digital Pin 11**
- Common cathode (or anode, depending on LED type) connected to **GND** (or **5V** respectively)
- The RGB LED is controlled by a second **potentiometer**:
  - **Middle pin** to **Analog Pin A0**
  - **Side pins** to **5V** and **GND**

Wiring Overview Summary

- The **joystick** adjusts the angles of two servo motors in real-time.
- A **potentiometer** modifies the brightness of a single **white LED**.
- A second **potentiometer** controls the color blending of an **RGB LED**.
- The circuit was assembled without the use of any current-limiting resistors.

Prototype



Conclusion

This project successfully implemented a simplified version of an endoscopic system using accessible components. Key functionalities included:

- Mechanical articulation through servos.
- Realistic and adjustable lighting.
- Camera module for image capture.

Future improvements could involve integrating wireless image transmission and mobile app visualization to further mimic advanced endoscopic systems.